Universidade Politécnica Bragança
Escola Superior Agrária

UTFPR
UNIVERSIDADE TECNOLÓGICA FEDERAL DO PARANÁ

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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Dissertation submitted to Escola Superior Agrária de Bragança to obtain the Degree of Master in Biotechnological Engineering under the scope of the double diploma with Universidade Tecnológica Federal do Paraná.

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This dissertation does not include the comments and suggestions mentioned by the Jury.

Bragança
November 2026

Acknowledgements

Abstract

The increasing generation of food waste (FW) and surplus of crude glycerol (C-GL) from biodiesel production represent two significant organic waste streams that require sustainable management solutions. This study evaluated the technical feasibility of a single-stage anaerobic co-digestion system treating FW and C-GL as co-substrates, with the objective of verifying biogas production while providing a valorisation route for C-GL. The investigation was conducted in a 1 L continuous stirred-tank reactor (CSTR) operated under mesophilic conditions (35 °C) with an hydraulic retention time (HRT) of 30 days. Food waste was collected from a university canteen, and C-GL was obtained from biodiesel production residues; sludge from an anaerobic reactor from a municipal wastewater treatment plant (WWTP) served as the inoculum. The reactor was operated in sequential phases: 1-an initial adaptation period, 2-co-digestion of FW and C-GL, 3-C-GL based feeding, and 4-a post-feeding period. CH_4 production was monitored continuously using Automatic Methane Potential Test System (AMPTS II), while process stability was assessed through measurements of pH, total Alk (TA), volatile fatty acid (VFA), and their VFA/TA. Substrate characterisation included total solids (TS), volatile solids (VS), density, total carbon (TC), and total nitrogen (TN) analyses. The results demonstrated that phase 2 (co-digestion of FW and C-GL) yielded the highest CH_4 production rates and cumulative volumes, with flow spikes immediately following feeding events indicating rapid substrate conversion. In contrast, phase 3 (C-GL based feeding) resulted in reduced biogas production despite continued methanogenic activity. Total alkalinity (Alk) remained elevated following bicarbonate supplementation, maintaining pH around 7 throughout most of the experiment; VFA concentrations fluctuated between 3,000 mg CH_3COOH L^{-1} to 5,000 mg CH_3COOH L^{-1}, with VFA/TA of 0.3 to 1.2. These findings support the hypothesis that C-GL can be useful for biogas production when co-digested with FW or alone. The study demonstrated the technical viability of anaerobic co-digestion as a strategy for valorising C-GL within a circular bioeconomy framework, though further research is needed to optimise C-GL loading rates and develop robust feeding strategies for industrial

application.

Keywords: anaerobic digestion; biogas; crude glycerol; food waste; methane.

Resumo

A crescente geração de resíduos alimentares (RA) e o excedente de glicerol bruto (GL-B) proveniente da produção de biodiesel representam dois fluxos significativos de resíduos orgânicos que requerem soluções de gestão sustentáveis. Este estudo avaliou a viabilidade técnica de um sistema de codigestão anaeróbia de uma única etapa, tratando RA e GL-B como co-substratos, com o objetivo de verificar a produção de biogás e, simultaneamente, proporcionar uma via de valorização para o GL-B. A investigação foi conduzida num reator de tanque agitado contínuo (CSTR) de 1 L, operado em condições mesofílicas (35 °C), com um tempo de retenção hidráulica (TRH) de 30 dias. Resíduos alimentares foram recolhidos numa cantina universitária, e o GL-B foi obtido a partir de resíduos da produção de biodiesel; lamas de um reator anaeróbio de uma estação de tratamento de águas residuais (ETAR) municipal serviram como inóculo. O reator foi operado em fases sequenciais: 1–um período inicial de adaptação; 2–codigestão de RA e GL-B; 3–alimentação baseada em GL-B; e 4–um período pós-alimentação. A produção de CH_4 foi monitorizada continuamente utilizando *Automatic Methane Potential Test System* (AMPTS II), enquanto a estabilidade do processo foi avaliada através de medições de pH, Alc total (AT), ácido gordo volátil (AGV) e respectiva AGV/AT. A caracterização dos substratos incluiu análises de sólidos totais (ST), sólidos voláteis (SV), massa volúmica, carbono total (CT) e nitrogénio total (NT). Os resultados demonstraram que a Fase 2 (codigestão de RA e GL-B) originou as taxas de produção de CH_4 e os volumes acumulados mais elevados, com picos de fluxo imediatamente após as alimentações, indicando uma rápida conversão do substrato. Em contraste, a Fase 3 (alimentação baseada em GL-B) resultou numa produção reduzida de biogás, apesar da atividade metanogénica continuada. A AT manteve-se elevada após a suplementação com bicarbonato, mantendo o pH em torno de 7 durante a maior parte da experiência; as concentrações de AGV flutuaram entre 3.000 mg CH_3COOH L^{-1} a 5.000 mg CH_3COOH L^{-1}, com AGV/AT de 0,3 a 1,2. Estes resultados apoiam a hipótese de que o GL-B pode ser útil para a produção de biogás quando codigerido com RA ou isoladamente. O estudo demonstra a viabilidade técnica da codigestão anaeróbia como

estratégia de valorização do GL-B no âmbito de uma bioeconomia circular, embora seja necessária investigação adicional para otimizar as taxas de carga de GL-B e desenvolver estratégias de alimentação robustas para aplicação industrial.

Palavras-chave: biogás; digestão anaeróbia; glicerol bruto; metano; resíduos alimentares.

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List of Abbreviations

AD	anaerobic digestion
Alk	alkalinity
AMPTS II	Automatic Methane Potential Test System
C-GL	crude glycerol
CSTR	continuous stirred-tank reactor
FS	fixed solids
FW	food waste
GL	glycerol
HRT	hydraulic retention time
MONG	matter organic non-glycerol
OLR	organic loading rate
P-GL	pure glycerol
TA	total Alk
TC	total carbon
TN	total nitrogen
TOC	total organic carbon
TS	total solids
UPB	Universidade Politécnica de Bragança
VFA	volatile fatty acid
VS	volatile solids
WWTP	wastewater treatment plant

List of Symbols

$Ca(HCO_3)_2$	Calcium bicarbonate
CH_4	Methane
C/N	Carbon-to-nitrogen ratio of the feed
CO_2	Carbon dioxide
$C_6H_{12}O_6$	Glucose
$KHCO_3$	Potassium bicarbonate
O_2	Oxigen
1,3-PDO	1,3-propanediol
VFA/TA	Ratio of volatile fatty acids to total alkalinity
v/v	Volume per Volume

List of Units

M	mega prefix (10^6)
k	kilo prefix (10^3)
m	milli prefix (10^{-3})
tonnes	metric tonne (1.000×10^3 kg)
g	gram
L	litre
m	metre
%	percentage
°C	degrees Celsius
min	minute
h	hour
d	day
yr	year
X	dilution factor (*X* times)
NmL	normalised millilitre (at standard conditions)
v	volume
mol dm^{-3}	molarity
G	relative centrifugal force

Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is food waste (FW), a global stream that in the European Union alone is estimated at 88 Mtonnes yr^{-1}, excluding inedible parts removed from the supply chain (Scherhaufer *et al.*, 2018). When considering only post-processing activities (wholesale, retail, food service, and households), approximately 66.3 Mtonnes of FW is produced annually in the EU-27+1 (Albizzati; Tonini; Astrup, 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of FW is substantial. It accounts for approximately 15 % to 16 % of the total climate impact of the entire food supply chain, translating to approximately 186 Mtonnes CO_2-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhaufer *et al.*, 2018). Most of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favored over landfill disposal (Albizzati; Tonini; Astrup, 2021; Scherhaufer *et al.*, 2018). Despite this, much FW continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

In parallel to the FW challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: crude glycerol (C-GL). Formed during the transesterification of triglycerides, approximately 10 kg of C-GL is produced for every 100 kg of biodiesel (Garlapati; Shankar; Budhiraja, 2016). Global biodiesel output, driven by renewable energy mandates, has caused the glycerol (GL) market to swell with C-GL supply projected to reach about 6.33 Mtonnes by 2025, of which nearly 680 ktonnes are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachi;

Kingsley; Spallina, 2023).

Crude glycerol composition varies widely depending on the oil source and catalyst used; however, it typically contains methanol, soaps, free fatty acids, salts and other matter organic non-glycerol (MONG) compounds (Anand; Saxena, 2012; Attarbachi; Kingsley; Spallina, 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-PDO production when untreated C-GL from jatropha, soybean or sunflower oils was used (Anand; Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachi; Kingsley; Spallina, 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, C-GL has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher *et al.*, 2025; Tomczak *et al.*, 2025).

Anaerobic digestion is a mature biological process in which consortia of microorganisms convert organic matter into biogas, primarily CH_4 (55 % to 65 %) and CO_2, under O_2-free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis, and methanogenesis (Gebreegziabher *et al.*, 2025). Anaerobic digestion offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertilizer; and reduces the environmental load of organic waste by avoiding landfilling and its associated CH_4 emissions (Scherhaufer *et al.*, 2018; Gebreegziabher *et al.*, 2025).

Within this framework, anaerobic co-digestion, the simultaneous treatment of two or more substrates, has emerged as a particularly effective strategy. Co-digesting C-GL with nitrogen-rich materials, such as animal manure, sewage sludge, or FW, balances the C/N, dilutes potential inhibitors, provides essential macro- and micro-nutrients, and fosters synergistic microbial interactions (Tomczak *et al.*, 2025; Gebreegziabher *et al.*, 2025). The results can be a dramatic improvement in biogas and CH_4 yields: studies report increases ranging from 14 % to 226 % when C-GL is added to a variety of base substrates, with CH_4 contents generally exceeding 60 % as long as GL does not exceed approximately 8 % (v/v) of the feed (Gebreegziabher *et al.*, 2025; Tomczak *et al.*, 2025).

However, excessive GL loads can lead to volatile fatty acid (VFA) accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher *et al.*, 2025; Tomczak *et al.*, 2025). Overall, the convergence of abundant FW and surplus C-GL supplies, together with the proven capability of anaerobic digestion (AD) to

convert these streams into renewable CH_4 and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this study was to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic FW and C-GL as substrates. The investigation was conducted by determining biogas production, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of C-GL, a by-product of the biodiesel industry, helps biogas production because of its high energy content, provided that adequate operational conditions are maintained to avoid VFA overload.

2.2 Specific Objectives

1. Characterize the substrates (FW and C-GL) and evaluate the physicochemical parameters relevant to anaerobic digestion.
2. Operate the single-stage anaerobic co-digestion reactor and assess process stability.
3. Efficiency evaluation of the process for the utilization of different substrates and organic loading rates (OLRs).

Chapter 3

Literature Review

The rapid expansion of biodiesel production has generated an oversupply of C-GL, a by-product of the transesterification. Fig. 3.1 shows the steps to produce biodiesel and when GL is produced (Garlapati; Shankar; Budhiraja, 2016). This waste stream, loaded with impurities such as methanol, soaps, and hydroxides. Substances that, poses environmental disposal challenges. Also, with the variation of impurities depending on the source, there is no standard method to use this source of carbon (Anand; Saxena, 2012; Gebreegziabher *et al.*, 2025). Fig. 3.2 shows the processes and substances that contaminate GL. Anaerobic digestion aligns with circular economy principles by converting organic residues into biogas while recycling nutrients and making C-GL a promising co-substrate (Gebreegziabher *et al.*, 2025).

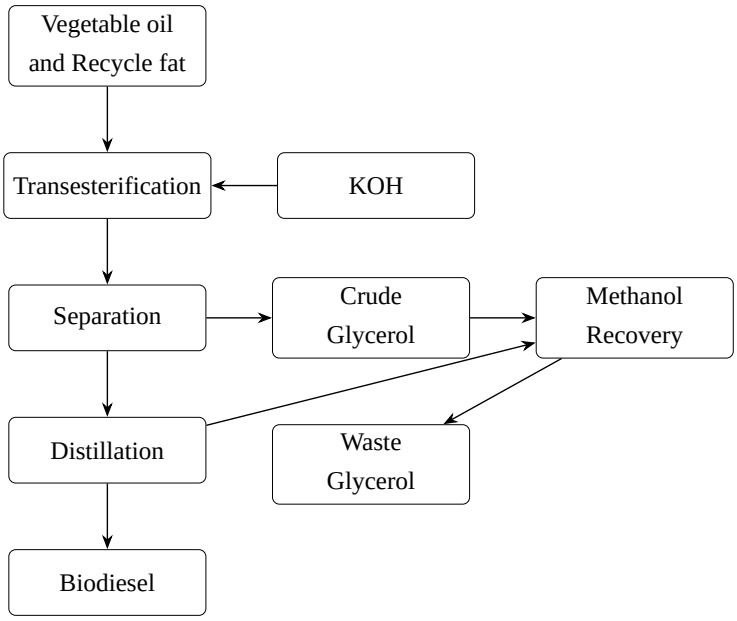


Figure 3.1: Crude glycerol from biodisel production (Adapted from: Garlapati, Shankar, and Budhiraja (2016))

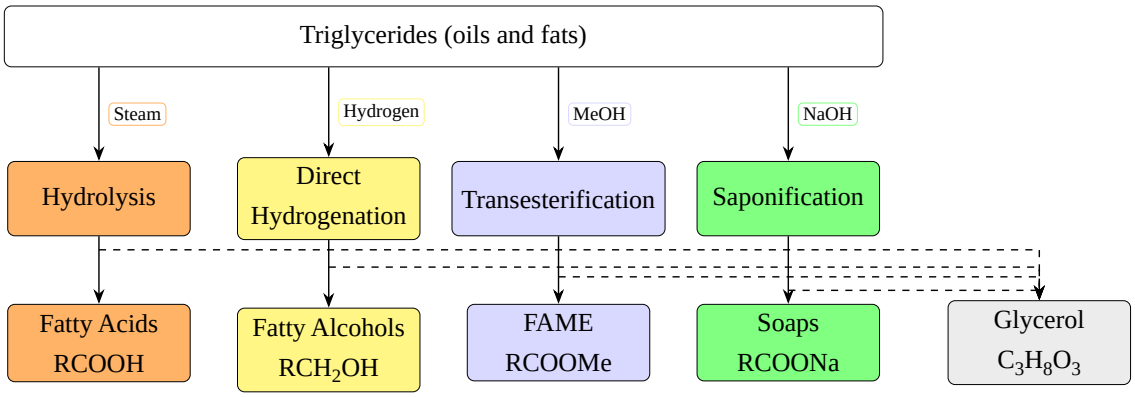


Figure 3.2: Contaminations of the crude glycerol (Adapted from: Attarbachi, Kingsley, and Spallina (2023))

Gebreegziabher *et al.* (2025) reported that co-digestion of C-GL can boost CH_4 production compared to mono-digestion of the base substrate. Tomczak *et al.* (2025) reviewed one-stage AD experiments and concluded that a CH_4 content above 60 % could be consistently obtained when the C-GL fraction did not exceed 8 % (v/v) of the feed, as shown in the flow chart by Li, Park, and Zhu (2011) in Fig. 3.3 where CH_4 production could be up to 70 %. However, the same review emphasized that increasing the GL loading beyond a system-specific threshold may provoke inhibitory effects, likely due to the accumulation of

VFAs and impurities present in C-GL (Anand; Saxena, 2012; Tomczak *et al.*, 2025). The heterogeneous composition of C-GL, affected by oil feedstock, catalyst, and biodiesel processing, demands careful feedstock characterisation and, if necessary, simple pretreatment to avoid process upsets (Attarbachi; Kingsley; Spallina, 2023).

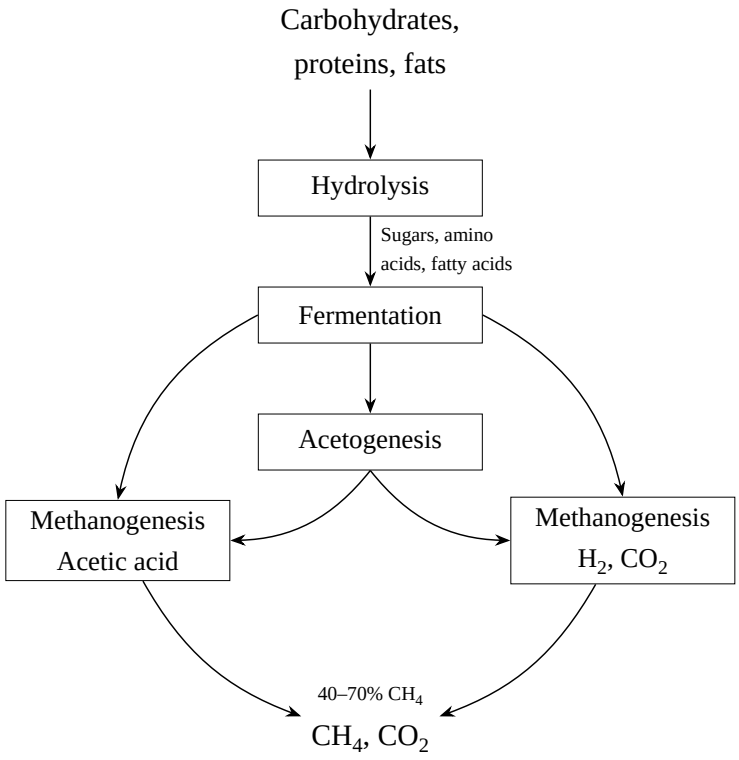


Figure 3.3: CH_4 production from anaerobic digestion (Adapted from: Li, Park, and Zhu (2011))

Liquid AD, typically conducted in a continuous stirred-tank reactor (CSTR), operates at total solid concentrations of 0.5 % to 15 % and is generally used for treating sewage sludge, manure, and FW (Li; Park; Zhu, 2011). The CSTR provides uniform distribution of substrates, microorganisms, and heat, which is essential when feeding a fermentable substrate. Temperature is a critical operational parameter, as the activity of methanogenic consortia is strongly temperature-dependent, and both mesophilic and thermophilic regimes have been applied to AD (Choorit; Wisarnwan, 2007). Tomczak *et al.* (2025) noted that the majority of GL co-digestion studies have been performed under mesophilic conditions, while thermophilic operation, which could offer higher conversion rates, still requires more systematic investigation. The CSTR platform is well suited to explore such temperature effects because its design allows easy control of heating and hydraulic retention time.

The selection of a robust microbial inoculum is crucial for rapid start-up and stable performance of a CSTR fed with GL. Anaerobic sludge from municipal wastewater treatment plants (WWTPs) is the most common source of inoculum because it harbors a diverse consortium of hydrolytic, acidogenic, and methanogenic microorganisms. The flow chart from Gebreegziabher *et al.* (2025) on Fig. 3.4 shows each step of AD until CH_4 production. Posmanik *et al.* (2020) demonstrated that granular sludge, obtained from up-flow anaerobic sludge blanket reactors treating wastewater, outperforms suspended biomass as an inoculum. Although their work focused on batch tests, the findings strongly suggest that using granular sludge from a WWTP as seed for a CSTR could accelerate acclimation, improve process stability, and reduce the risk of acidification during load increases.

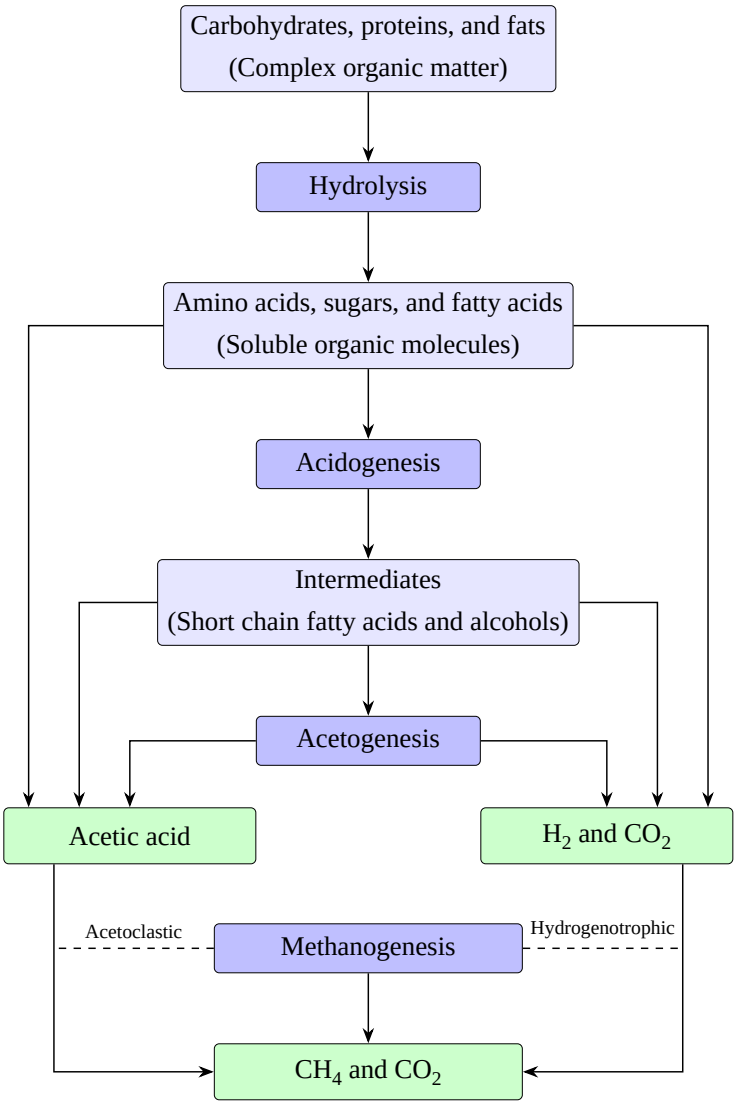


Figure 3.4: Steps to produce CH_4 from anaerobic digestion (Adapted from: Gebreegziabher *et al.* (2025))

When GL starts to degrade, it makes the AD process susceptible to VFA accumulation and pH drop if the organic loading rate is not properly controlled. Therefore, close monitoring of process indicators is essential. Ribas, Moraes, and Foresti (2007) evaluated several methods for determining these parameters specifically in anaerobic reactors, highlighting the importance of accurate VFA and alkalinity (Alk) measurements for timely detection of process imbalance. The method proposed by DiLallo and Albertson (DiLallo; Albertson, 1961 apud Ribas; Moraes; Foresti, 2007) provides a validated method for measuring VFAs and Alk. Maintaining a favorable VFA/TA ratio is particularly critical when co-digesting GL, as excess VFAs can inhibit methanogenesis (Tomczak *et al.*, 2025).

The available literature confirms that C-GL is a high-energy substrate that can significantly boost CH_4 production in liquid AD systems. A CSTR provides the mixing and temperature control required to harness this potential. Using granular sludge from a WWTP as the inoculum can enhance start-up speed and methanogenic activity, as demonstrated by comparative studies (Posmanik *et al.*, 2020; Li; Park; Zhu, 2011). To unlock the full benefits while avoiding inhibition, the GL loading must be kept below the documented thresholds, and the reactor state should be continuously tracked through standard measurements of VFAs and Alk. Further research is needed to optimize thermophilic CSTR operation and establish long-term performance data for granular-sludge-inoculated reactors treating C-GL.

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

Food waste was collected from the canteen at Universidade Politécnica de Bragança (UPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize sugar content and minimize nitrogen content. It was collected 3,235 g of waste, which was then diluted with approximately 2.5 L of distilled water and homogenized using a blender to obtain a FW suspension with 5,526 g total. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce *et al.*, 2020).

4.2 Crude Glycerol Collection

A 6 L bottle of C-GL, obtained from previous biodiesel production experiments at UPB, was used in this study. C-GL from biodiesel typically contains impurities, such as catalyst residues (Attarbachi; Kingsley; Spallina, 2023), which may influence its behavior during anaerobic digestion.

4.3 Inoculum Collection

The inoculum was collected from an anaerobic digester at the Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for anaerobic digestion (Posmanik *et al.*, 2020).

4.4 Physical and Chemical Characterisation

total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the FW, C-GL, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for FW and C-GL.

4.4.1 TS and VS Determination

Total solids (TS) and volatile solids (VS) were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Lipps; Braun-Howland; Baxter, 2023). The method involves controlled drying and calcination steps to quantify the TS and VS fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. After drying, the crucibles were cooled in desiccator at 105 °C for 1 h, the process and weighed again to determine the TS. After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Thereafter, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

Total solids, and volatile solids were calculated as follows:

$$\mathrm{TS} = \frac{M_{105} - M_{\mathrm{crucible}}}{V_{\mathrm{sample}}} \quad \left[\mathrm{g\,L^{-1}}\right] \tag{4.1}$$

$$\mathrm{VS} = \frac{M_{105} - M_{550}}{V_{\mathrm{sample}}} \quad \left[\mathrm{g\,L^{-1}}\right] \tag{4.2}$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, M_{crucible} is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in L and the mass is in g (Table 4.1).

Table 4.1: Total solids, Volatile solids and Fixed solids values

Sample		$M_{crucible}$ (g)	M_{wet} (g)	M_{sample} (g)	V_{sample} (L)	M_{105} (g)	TS (g L^{-1})	Average TS (g L^{-1})	M_{550} (g)	VS (g L^{-1})	Average VS (g L^{-1})
Food Waste	1	7.873e1	1.060e2	2.726e1	2.550e−2	9.100e1	4.810e2		7.920e1	4.625e2	
	2	8.432e1	1.133e2	2.898e1	2.710e−2	9.887e1	5.364e2	5.364e2	8.474e1	5.208e2	5.208e2
	3	7.759e1	1.088e2	3.125e1	2.920e−2	9.344e1	5.420e2		7.818e1	5.216e2	
Crude Glycerol	1	8.422e1	8.864e1	4.424	5.000e−3	8.850e1	8.634e2		8.421e1	8.644e2	
	2	7.348e1	7.799e1	4.509	5.000e−3	7.786e1	8.677e2	8.677e2	7.347e1	8.687e2	8.687e2
	3	8.620e1	9.061e1	4.410	4.900e−3	9.050e1	8.713e2		8.620e1	8.724e2	
Inoculum	1	8.991e1	9.745e1	7.537	7.600e−3	8.997e1	6.130e−1		8.992e1	4.620e−1	
	2	8.017e1	8.873e1	8.564	8.600e−3	8.022e1	5.244e−1	5.244e−1	8.018e1	3.643e−1	3.643e−1
	3	7.843e1	8.679e1	8.356	8.400e−3	7.848e1	4.663e−1		7.845e1	3.105e−1	

4.4.2 Density Determination

Density was determined using a gravimetric volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed again to verify the reading accuracy (density of water at $\rho_{20\,°C} = 998.21\,\mathrm{g\,L^{-1}}$). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \quad \left[\mathrm{g\,L^{-1}}\right] \tag{4.3}$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume, the masses were measured in g and the volume in L (Table 4.2).

Table 4.2: Results of density for substrates and inoculum

Sample		M_{cylinder} (g)	M_{total} (g)	V_{sample} (L)	Density ($\mathrm{g\,L^{-1}}$)	Average Density ($\mathrm{g\,L^{-1}}$)
Food	1	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
Waste	2	1.160e2	2.230e2	1.002e−1	1.068e3	
Crude	1	1.142e2	2.035e2	9.990e−2	8.932e2	8.931e2
Glycerol	2	1.157e2	2.044e2	9.930e−2	8.929e2	
Inoculum	1	1.325e2	2.314e2	9.920e−2	9.978e2	9.950e2
	2	1.157e2	2.150e2	1.000e−1	9.922e2	

4.4.3 Determination of TOC and TN

Total organic carbon (TOC), total carbon (TC) and TN were measured using a Shimadzu TOC-5000A analyzer. Prior to analysis, the FW was diluted 400 X, and the C-GL was diluted 800 X to bring the concentrations within the linear range of the instrument. The results are on the Table 4.3.

Table 4.3: Total carbon and Total nitrogen for substrates

Substrate		TOC ($\mathrm{mg\,L^{-1}}$)	TC ($\mathrm{mg\,L^{-1}}$)	Average TC ($\mathrm{mg\,L^{-1}}$)	TN ($\mathrm{mg\,L^{-1}}$)	Average TN ($\mathrm{mg\,L^{-1}}$)
Food	1	5.039	2.018e3	1.779e3	2.756e2	2.818e2
Waste	2	3.927	1.540e3		2.879e2	

Table 4.3 continued

Substrate		TOC (mg L^{-1})	TC (mg L^{-1})	Average TC (mg L^{-1})	TN (mg L^{-1})	Average TN (mg L^{-1})
Crude Glycerol	1	8.808	7.050e3	5.523e3	4.700e2	4.886e2
	2	4.992	3.997e3		5.072e2	

4.5 Experimental Setup

The reactor had a working volume of 1 L and was operated in semi-continuous mode, with feeding performed once daily, under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow outlet was kept closed with a valve to maintain an anaerobic environment. The experimental setup is shown in Fig. 4.1:

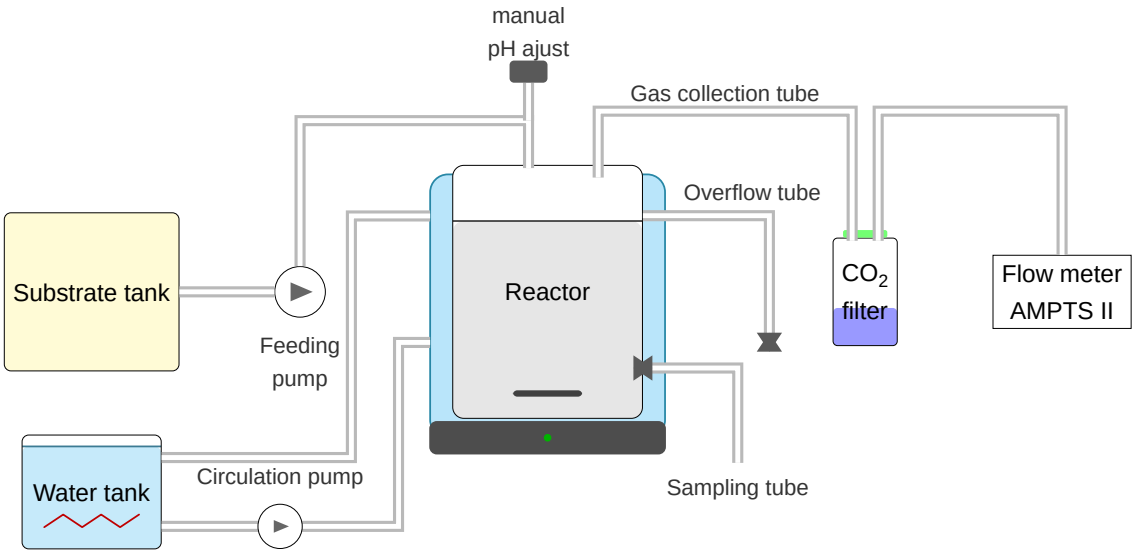


Figure 4.1: Schematic of the experimental setup used

The reactor was maintained at 35 °C, which is close to the optimal temperature for mesophilic microorganisms (Choorit; Wisarnwan, 2007). The OLR was set at a maximum of 4 kg VS m^{-3} d^{-1}, with an hydraulic retention time (HRT) of 30 days, which is suitable for stable co-digestion of FW and GL (Tomczak *et al.*, 2025). The pH was maintained at approximately 7 by adding $KHCO_3$ to the feed solution at a concentration of 10 g L^{-1}, and to the reactor whenever the pH dropped below 6.0.

In the Eq. (4.4) and (4.5), about HRT and OLR, respectively:

$$\mathrm{HRT} = \frac{V}{Q} \quad [\mathrm{d}] \tag{4.4}$$

$$\mathrm{OLR} = \frac{S_{\mathrm{in}} \times Q}{V} \quad \left[\mathrm{kg\ VS\ m^{-3}\ d^{-1}}\right] \tag{4.5}$$

where S_{in} is the influent VS concentration (kg $\mathrm{m^{-3}}$), Q is the volumetric flow rate ($\mathrm{m^3\ d^{-1}}$), and V is the reactor working volume ($\mathrm{m^3}$).

4.5.1 Feeding Solution

The feed was designed to achieve a C/N ratio of 25:1, which is optimal for AD (Li; Park; Zhu, 2011). Characterisation of the FW and C-GL showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both. Therefore, pure glycerol (P-GL) was used as an additive to allow precise control of the C/N ratio. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid FW suspension through the feeding tube.

The feed was made with a restriction of needing to have less than 15 % of TS. The amount of OLR and $KHCO_3$ added was decided by the pH and CH_4 production of the reactor. If it was too acidic, a lower OLR with $KHCO_3$ was used; if it was at the right amount it was kept the same; and if it was too basic, a higher OLR without $KHCO_3$ was added. The feed storage always had some remaining feed, and the new one was added on top, diluting or concentrating the feed going to the reactor.

The feed was also supplemented with nutrients to improve the growth of microorganisms. Table 4.4 lists the nutrients added to the substrate and their respective concentrations.

Table 4.4: Nutrients added to the substrate and there dosages. (Adapted from: Alqudah (2025))

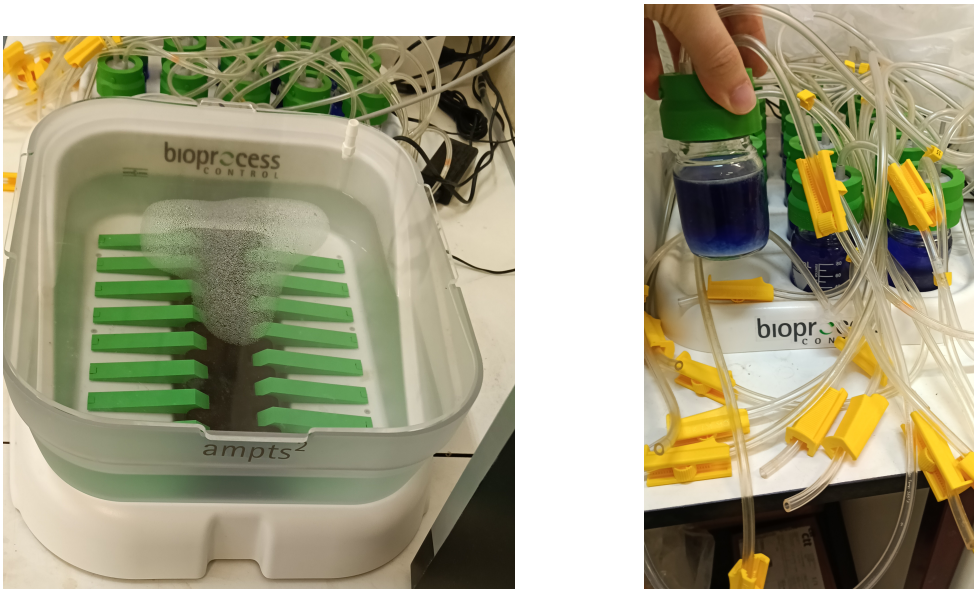
Nutrients	Concentration (mg $\mathrm{L^{-1}}$)
$MgSO_4 \cdot 7\,H_2O$	5.400e1
$FeSO_4 \cdot 7\,H_2O$	1.494e3
$CoCl_2 \cdot 6\,H_2O$	1.200
$ZnCl_2$	3.000e1
$NiCl_2 \cdot 6\,H_2O$	3.000e−2
EDTA	6.000e−1

Table 4.4 continued

Nutrients	Concentration (mg L^{-1})
$CuSO_4 \cdot 5\,H_2O$	2.637e1

4.5.2 Biogas Production

The Automatic Methane Potential Test System (AMPTS II) system (hardware Fig. 4.2, software Fig. 4.3) was used to measure the gas flow produced by the reactor. A CO_2-fixing unit with a solution of 3 mol dm^{-3} NaOH was placed before the flow meter, so the recorded values corresponds essentially to the volume and flow of CH_4.



(a) Automatic Methane Potential Test System gas flow measurer

(b) Automatic Methane Potential Test System CO_2 trap

Figure 4.2: Automatic Methane Potential Test System hardware

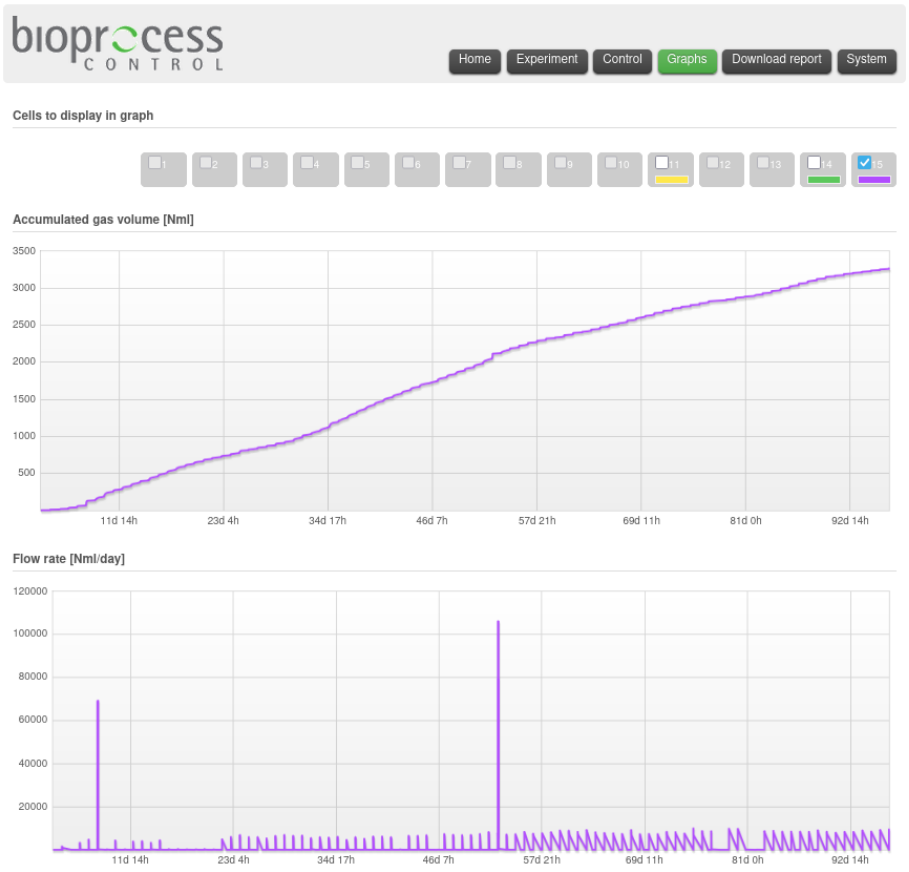


Figure 4.3: Automatic Methane Potential Test System software

4.5.3 Analysis Methods

The method of DiLallo and Albertson (DiLallo; Albertson, 1961 apud Ribas; Moraes; Foresti, 2007) was used to measure Alk and VFA. The collected sample was first centrifuged at 2,604 G for 30 minutes, to filter the particulates before performing the measurements.

The volumes were measured in mL, the sample volume was always 50 mL, the normality (N) of the acid and base, at first was 0.1, but then it was changed to 1.0. The Eq. (4.6) is expressed in mg $CaCO_3$ L^{-1}, and Eq. (4.7) is expressed in mg CH_3COOH L^{-1}.

$$\text{TA} = \frac{N_{\text{acid}} * V_{\text{acid}} * 50000}{V_{\text{sample}}} \quad \left[\text{mg } CaCO_3 \text{ } L^{-1}\right] \tag{4.6}$$

$$\text{VFA} = \frac{N_{\text{base}} * V_{\text{base}} * 60000}{V_{\text{sample}}} \quad \left[\text{mg } CH_3COOH \text{ } L^{-1}\right] \tag{4.7}$$

4.5.4 Feeding of the reactor

Table 4.5 shows all the substrates used to feed the reactor and when they were made.

Table 4.5: Feed used throughout the experimental period

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	Volume Made	OLR &TS%	GL%	FW%
0[a]	FW, 156 g	$C_6H_{12}O_6$, 1.9 g		$Ca(HCO_3)_2$, 5 g	500 mL	1.5 & 15		31
4	C-GL, 23.2 mL	P-GL,[b] 31.5 mL		$KHCO_3$, 1.9 g	250 mL	0.8 & 10	22	
11	FW, 12.54 g	C-GL, 9.02 mL	P-GL, 2.6 mL	$KHCO_3$, 1.57 g	250 mL	0.13 to 0.4 & 5 to 10[c]	5	5
18	FW, 10.88 g	C-GL, 2.89 mL	P-GL, 1.44 mL		250 mL	0.025,6 & 2	2	4
25	FW, 31.12 g	C-GL, 44 mL	P-GL, 29 mL		250 mL	1.4 & 10	29	12
32	FW, 31.01 g	C-GL, 44 mL	P-GL, 28.9 mL		250 mL	1.4 & 10	29	12
39	FW, 46.52 g	C-GL, 66 mL	P-GL, 43 mL	$KHCO_3$, 2.6 g	250 mL	3.1 & 15	44	19
46	FW, 50.07 g	C-GL, 66 mL	P-GL, 43 mL	$KHCO_3$, 2.67 g	250 mL	3.16 & 15	44	20
53[d]	C-GL, 55 mL	P-GL, 11.5 mL		$KHCO_3$, 5.38 g	500 mL	0.42 & 10	13	
60	C-GL, 55 mL	P-GL, 12 mL		$KHCO_3$, 5.18 g	500 mL[e]	0.42 & 10	13	
67	C-GL, 41 mL	P-GL, 9 mL		$KHCO_3$, 2.56 g	250 mL	0.95 & 15	20	
74	C-GL, 60 mL	P-GL, 12.5 mL		$KHCO_3$, 2.57 g	250 mL	1.37 & 15	29	

[a] It only started feeding on day 2.

[b] The solution of P-GL was diluted 100 X.

[c] The FW wasn't well mixtured this day.

[d] Change of substrate.

[e] There was a leak on the feeding tube.

4.5.5 Performance Calculations

The following equations were used to calculate the performance indicators reported in the Chapter 6.

$$Q = \frac{V_r}{\mathrm{HRT}} \quad \left[\mathrm{m^3\ d^{-1}}\right] \tag{4.8}$$

$$m_{\mathrm{VS,day}} = \mathrm{OLR} \cdot V_r \quad \left[\mathrm{g\ d^{-1}}\right] \tag{4.9}$$

$$M_{\mathrm{VS}}(n) = \sum_{i=0}^{n} m_{\mathrm{VS,day}}(i) \quad [\mathrm{g}] \tag{4.10}$$

$$Y(n) = \frac{V_{\mathrm{CH_4}}(n)}{M_{\mathrm{VS}}(n)} \quad \left[\mathrm{NmL\ CH_4\ g^{-1}\ VS_{fed}}\right] \tag{4.11}$$

$$r_{\mathrm{CH_4}}(n) = \frac{F_{\mathrm{CH_4}}(n)}{m_{\mathrm{VS,day}}(n)} \quad \left[\mathrm{mL\ CH_4\ g^{-1}\ VS_{fed}\ d^{-1}}\right] \tag{4.12}$$

$$\dot{n}_{\mathrm{CH_4}}(n) = \frac{F_{\mathrm{CH_4}}(n)}{V_m \cdot V_r} \quad \left[\mathrm{mol\ CH_4\ m^{-3}\ d^{-1}}\right] \tag{4.13}$$

$$\dot{V}_{\mathrm{CH_4}}(n) = \frac{F_{\mathrm{CH_4}}(n)}{1000 \cdot V_r} \quad \left[\mathrm{L\ CH_4\ V_r^{-1}\ d^{-1}}\right] \tag{4.14}$$

$$\Delta V_{\mathrm{CH_4}}(a,b) = V_{\mathrm{CH_4}}(b) - V_{\mathrm{CH_4}}(a-1) \quad [\mathrm{NmL}] \tag{4.15}$$

$$\Delta M_{\mathrm{VS}}(a,b) = M_{\mathrm{VS}}(b) - M_{\mathrm{VS}}(a-1) \quad [\mathrm{g}] \tag{4.16}$$

$$Y(a,b) = \frac{\Delta V_{\mathrm{CH_4}}(a,b)}{\Delta M_{\mathrm{VS}}(a,b)} \quad \left[\mathrm{NmL\ CH_4\ g^{-1}\ VS_{fed}}\right] \tag{4.17}$$

$$\overline{\dot{n}_{\mathrm{CH_4}}}(a,b) = \frac{1}{b-a+1} \sum_{n=a}^{b} \dot{n}_{\mathrm{CH_4}}(n) \quad \left[\mathrm{mol\ CH_4\ m^{-3}\ d^{-1}}\right] \tag{4.18}$$

$$\overline{\dot{V}_{\mathrm{CH_4}}}(a,b) = \frac{1}{b-a+1}\sum_{n=a}^{b}\dot{V}_{\mathrm{CH_4}}(n) \quad \left[\mathrm{L\ CH_4\ V_r^{-1}\ d^{-1}}\right] \tag{4.19}$$

$$\overline{r_{\mathrm{CH_4}}}(a,b) = \frac{1}{b-a+1}\sum_{n=a}^{b}r_{\mathrm{CH_4}}(n) \quad \left[\mathrm{mL\ CH_4\ g^{-1}\ VS_{fed}\ d^{-1}}\right] \tag{4.20}$$

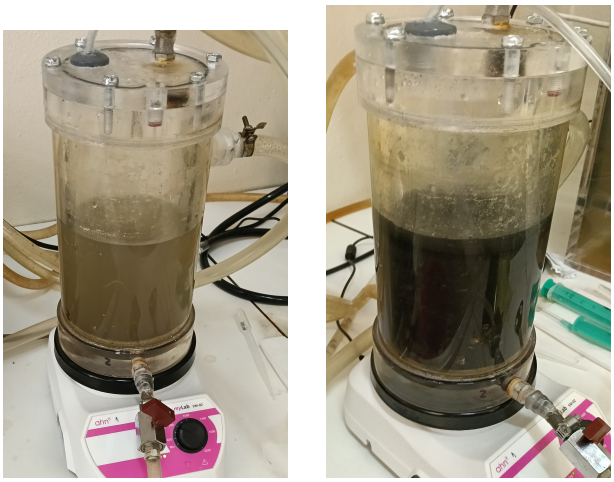
$$Y_{\mathrm{overall}} = \frac{V_{\mathrm{CH_4}}(N_f)}{M_{\mathrm{VS}}(N_f)} \quad \left[\mathrm{NmL\ CH_4\ g^{-1}\ VS_{fed}}\right] \tag{4.21}$$

Chapter 5

Results

5.1 Reactor Condition and Visual Observations

The reactor was inoculated with anaerobic sludge and initially contained a murky, light-brown suspension (Fig. 5.1a). Within three days of operation, the color changed to black (Fig. 5.1b), which is characteristic of active anaerobic digestion and is typically caused by the precipitation of metal sulfides, particularly iron sulfide, under reducing conditions. This visual change indicated the establishment of a suitable anaerobic environment and the onset of methanogenic activity.



(a) Murky suspension **(b)** Black suspention

Figure 5.1: Visual changes in the reactor, within the first week.

At the end of the experimental period, after approximately two weeks of unattended operation, four discrete masses resembling fungal colonies were observed on the internal

surfaces of the reactor during cleaning (Figs. 5.2a and 5.2b). The origin of these growths is uncertain, but they may have been introduced via the FW feed or during periods when the feeding tube was opened for repairs (e.g. leaks) and when changing the feed, while cleaning the tube. The presence of such organisms could have competed with the anaerobic microbial community for substrate, potentially affecting biogas production in the later stages of the experiment.



(a) Masses inside the dry reactor

(b) Masses floating in the reactor

Figure 5.2: Visual for the masses inside the reactor.

5.2 Biogas Production

Biogas production was continuously monitored using AMPTS II, which recorded both the cumulative volume of CH_4 produced and the instantaneous flow rate. The entire experimental period is summarised in Figs. 5.3 and 5.4. To facilitate interpretation, the data were divided into four distinct phases based on the feeding regime: (i) an adaptation phase 5.2.2, (ii) co-digestion of food waste and glycerol 5.2.3, (iii) digestion of crude glycerol 5.2.4, and (iv) a post-feeding period 5.2.5.

5.2.1 Overall Production

The cumulative CH_4 production over the entire experiment (Fig. 5.3) showed a continuous increase, indicating that the reactor maintained methanogenic activity throughout the monitored period. The total flow profile (Fig. 5.4) exhibited distinct peaks corresponding to the daily production of CH_4, with variations in amplitude reflecting the different substrate

compositions and organic loading rates applied during each phase. An outlier peak was observed on day 53, when the feed was changed from a mixture of FW and C-GL to only C-GL, and due to some instrumental complications, created an anomaly in the graph.

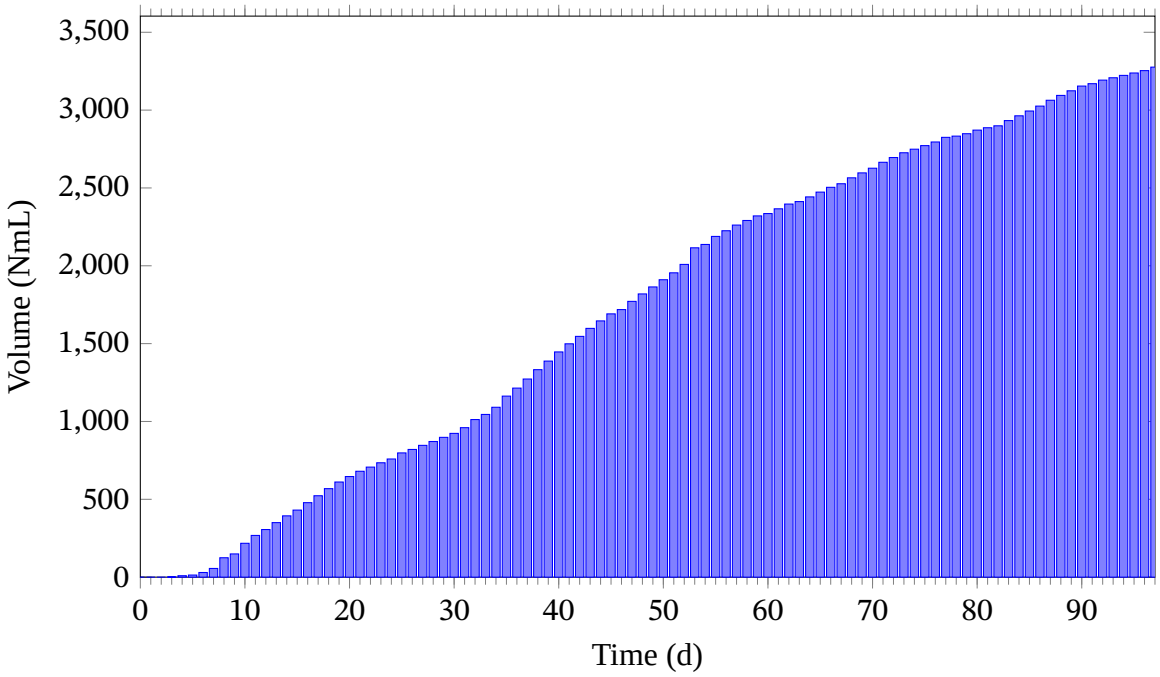


Figure 5.3: Cumulative volume of CH_4 NmL

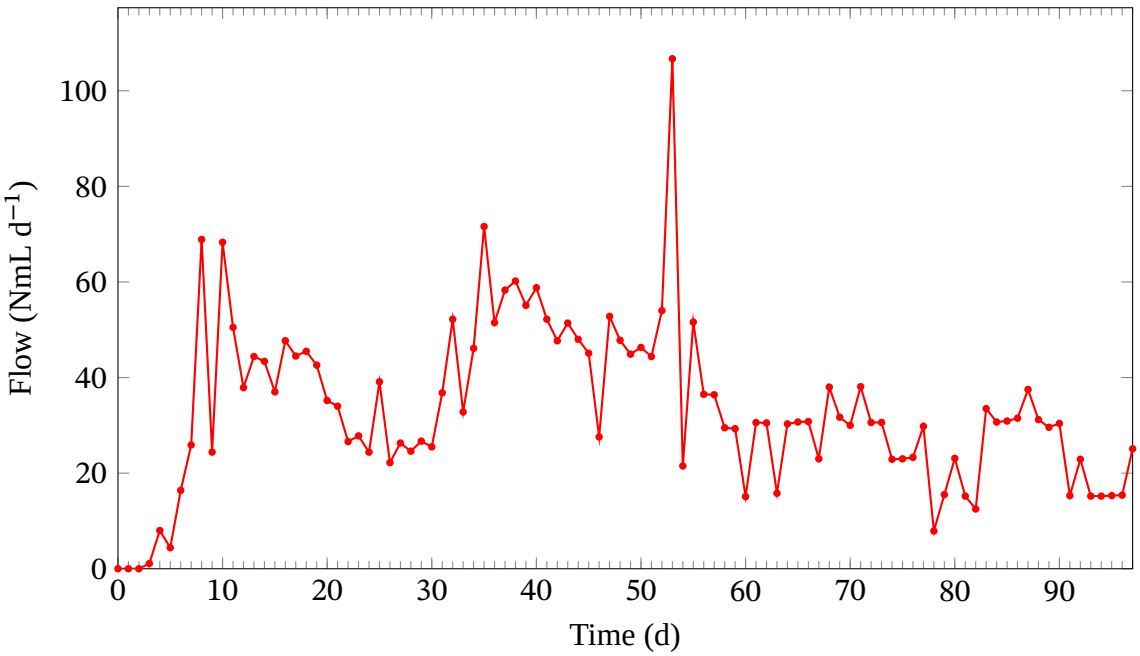


Figure 5.4: Volume flow of CH_4 NmL d^{-1}

5.2.2 Adaptation Phase

During the initial adaptation phase, CH_4 production was relatively low (Figs. 5.5 and 5.6). This lag period is normal for anaerobic reactors, as the microbial community adjusts to the new substrate and operating conditions. The small but measurable CH_4 flow confirmed the viability of the inoculum, and the gradual increase in flow suggested the onset of exponential growth of the microorganisms.

The spikes seen on days 8 and 10 (Fig. 5.6) probably occurred due to the first addition of C-GL to the feed.

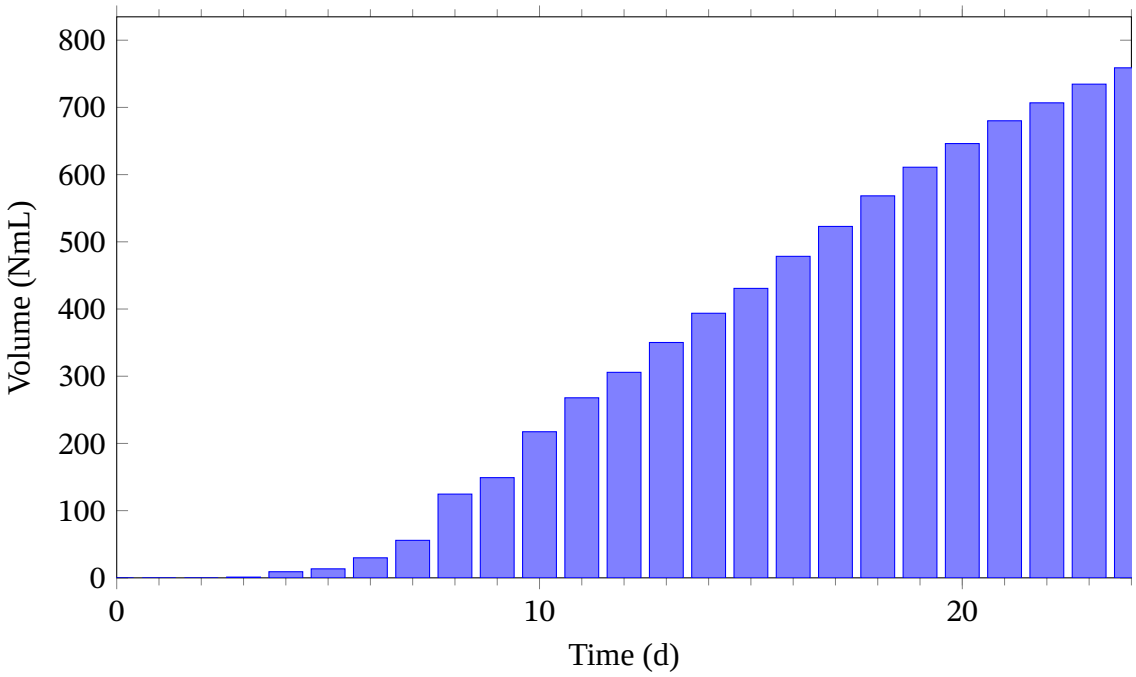


Figure 5.5: Cumulative volume of CH_4 NmL (Adaptation Phase)

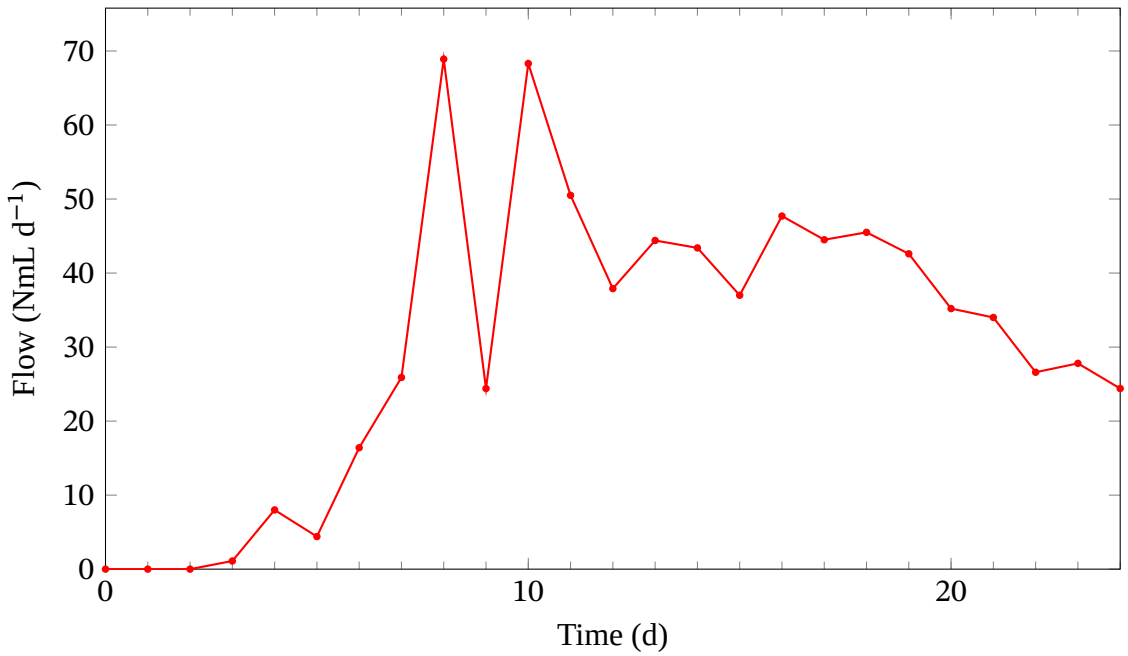


Figure 5.6: Volume flow of CH_4 NmL d^{-1} (Adaptation Phase)

5.2.3 Co-digestion of Food Waste and Crude Glycerol

When the reactor was fed with a mixture of FW and C-GL, a marked increase in CH_4 production was observed (Figs. 5.7 and 5.8). The cumulative CH_4 volume increased sharply, and the flow rate showed pronounced spikes immediately after each feeding event (Fig. 5.9). These spikes are typical of readily degradable substrates, which are quickly hydrolysed and converted to VFA and subsequently to CH_4. The average flow rate during this phase was higher than that in any other phase, indicating that the co-digestion of FW and C-GL provided a balanced nutrient supply and avoided the inhibition often associated with high GL concentrations alone.

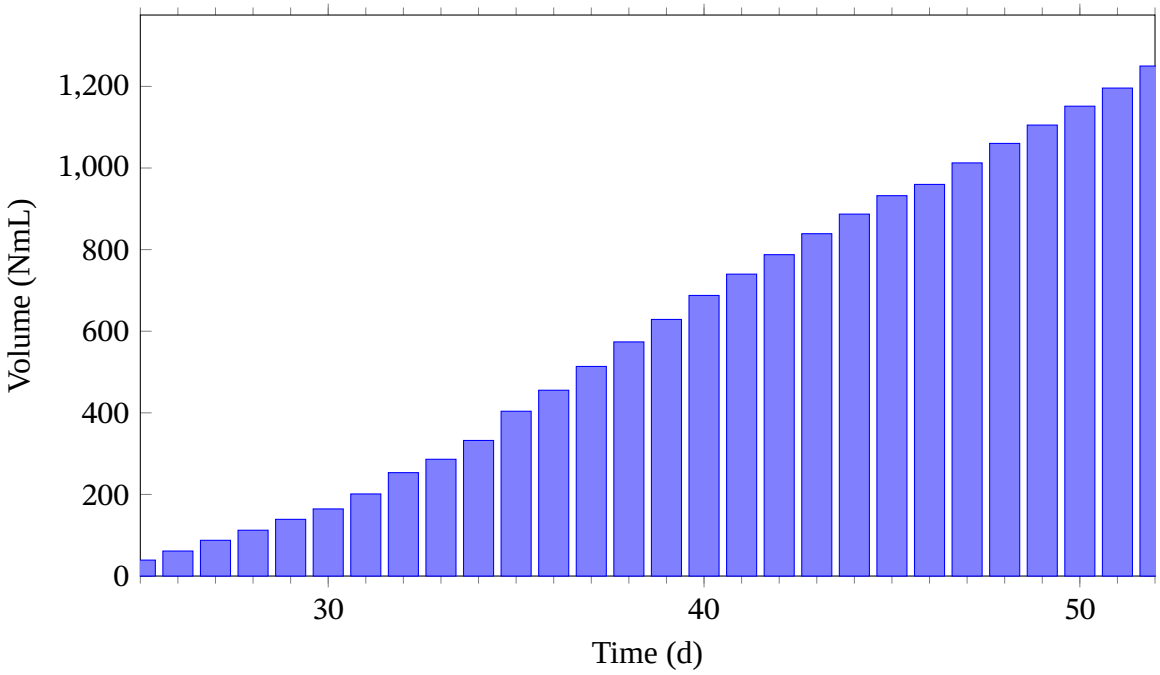


Figure 5.7: Cumulative volume of CH_4 NmL (co-digestion of food waste and crude glycerol)

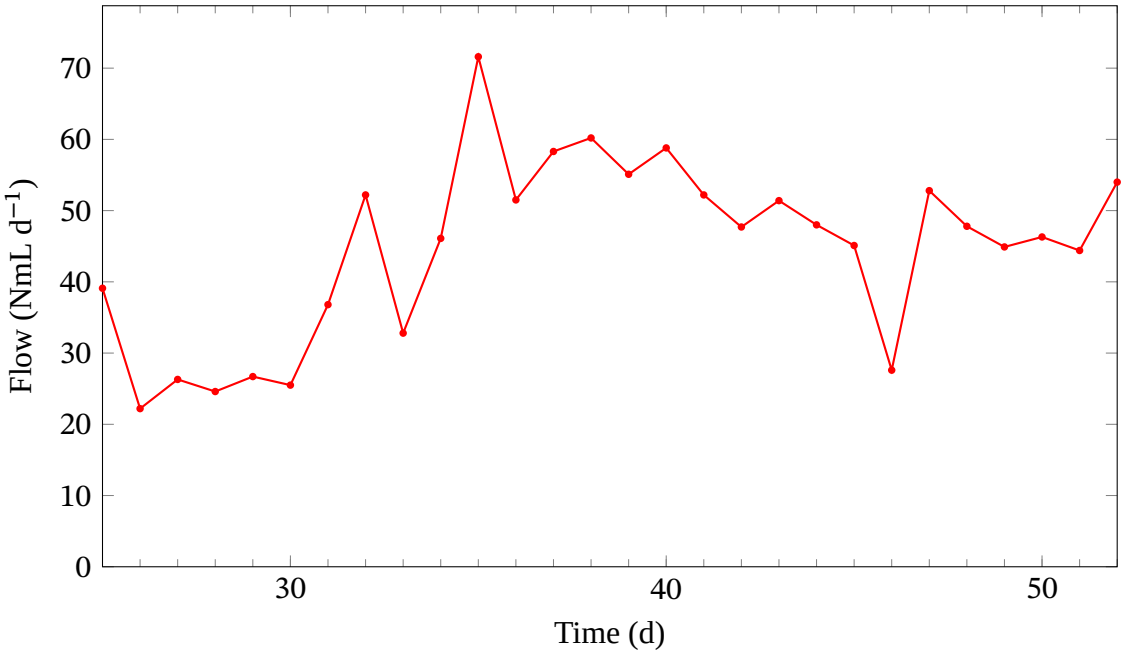


Figure 5.8: Volume flow of CH_4 NmL d^{-1} (co-digestion of food waste and crude glycerol)

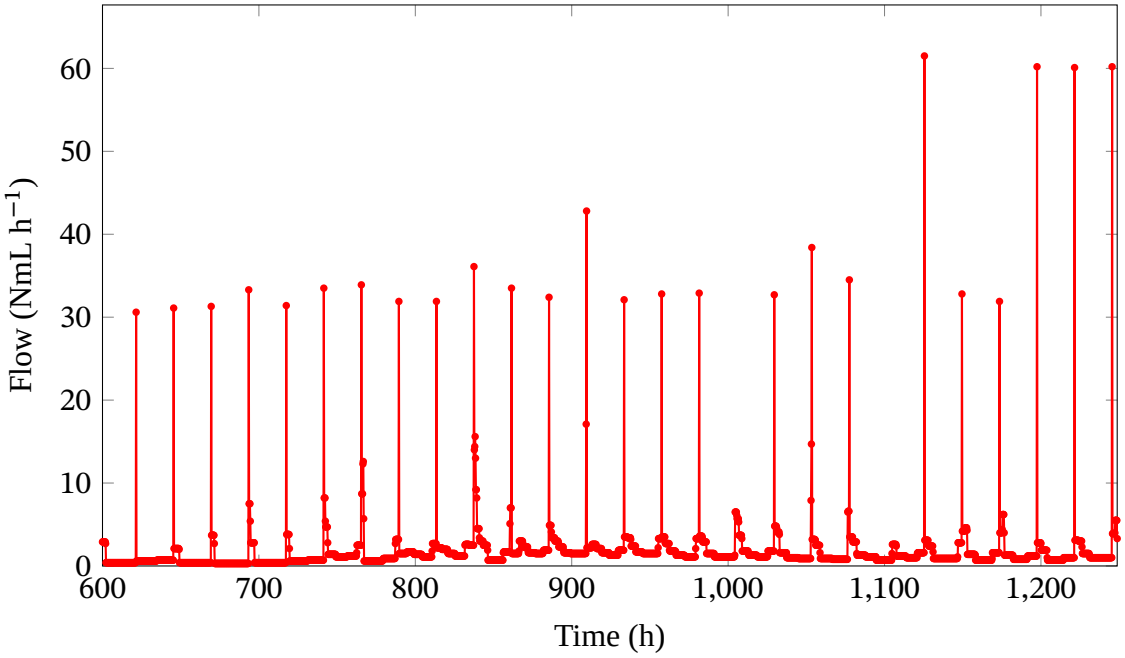


Figure 5.9: Volume flow of CH_4 NmL h^{-1} (co-digestion of food waste and crude glycerol)

5.2.4 Digestion of Crude Glycerol

When the feed was changed to C-GL alone, CH_4 production decreased (Figs. 5.10 and 5.11). Although spikes in the flow were still observed after feeding (Fig. 5.12), their magnitude was smaller than during the co-digestion phase, and the cumulative CH_4 volume plateaued more quickly. Several factors may explain this lower performance:

- The GL could have been transformed too fast or too slow, to maintain a concentration of intermediate products.
- The dilution rate of $\frac{1}{30}\,d^{-1}$ may have been insufficient to wash out inhibitory by-products or to maintain a stable population of glycerol fermenting bacteria.
- The appearance of fungal masses towards the end of this phase could have further stressed the anaerobic community by competing for substrate or introducing toxic metabolites. The fungal masses could have helped digest more complex sugars; therefore without a more thorough investigation of what it was, it can not be said it did not help the AD system.

Because the operational phases were performed sequentially in a single reactor, the direct effect of substrate composition cannot be separated from temporal effects, microbial adaptation, changes in OLR, and the previous operational history of the reactor. Therefore, the higher CH_4 production observed during the co-digestion phase should not be interpreted as definitive evidence of superior performance of this feeding strategy. A replicated experimental design with randomised phase order, or parallel reactors fed simultaneously with different substrates, would be required to isolate the effect of substrate composition from this effects.

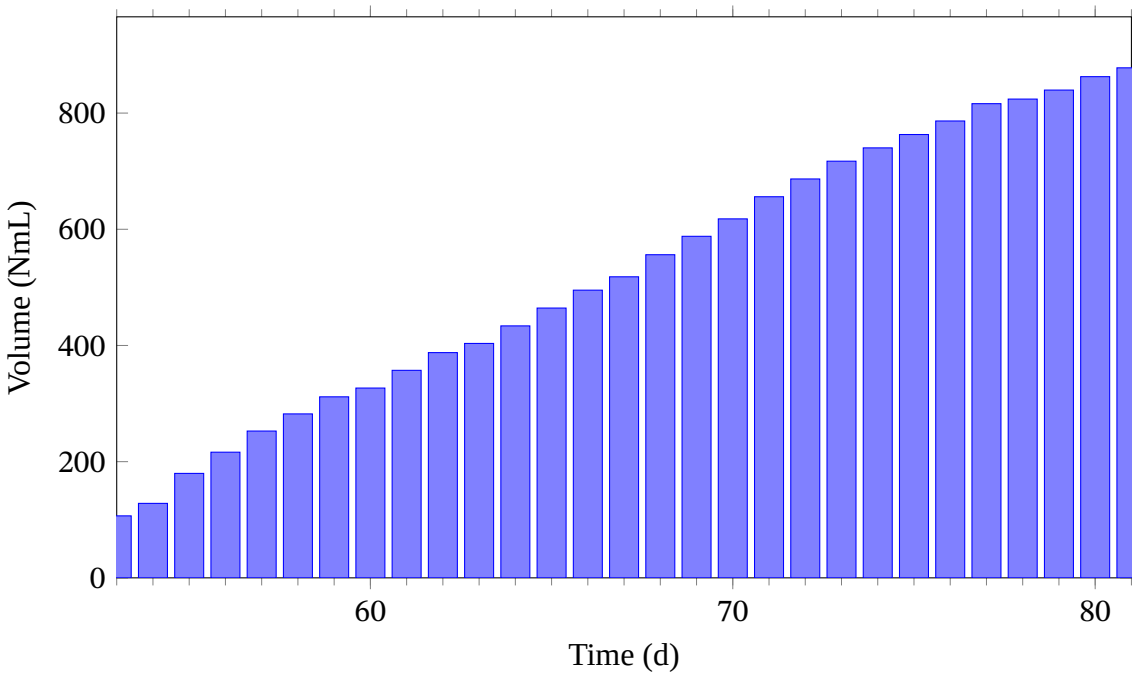


Figure 5.10: Cumulative volume of CH_4 NmL (Anaerobic digestion of crude glycerol)

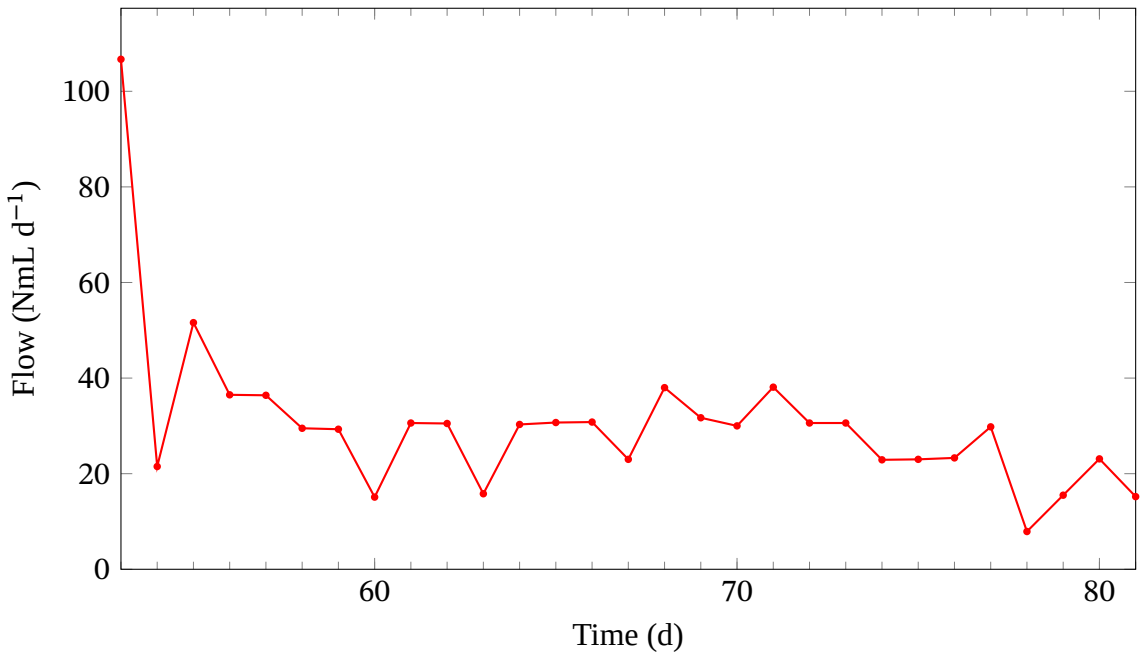


Figure 5.11: Volume flow of CH_4 NmL d^{-1} (Anaerobic digestion of crude glycerol)

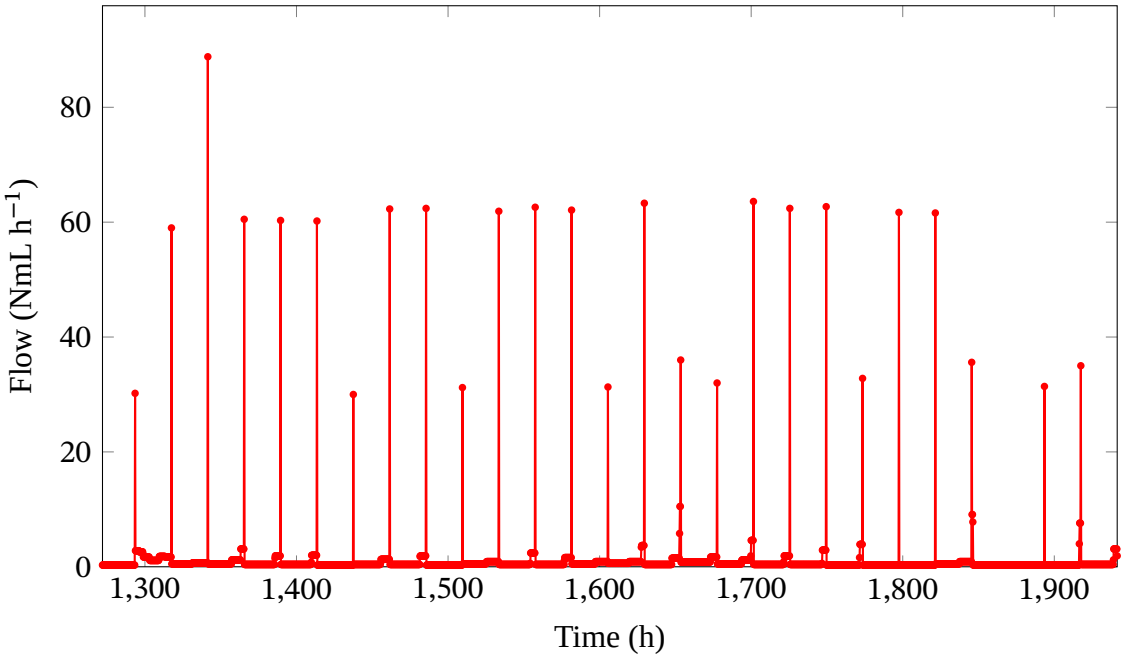


Figure 5.12: Volume flow of CH_4 NmL h^{-1} (Anaerobic digestion of crude glycerol)

5.2.5 Post-feeding Period

During the final two weeks, no fresh substrate was added to the reactor; it only had some residual substrate from the C-GL based phase for a week. Nevertheless, a small residual CH_4 production was recorded (Figs. 5.13 and 5.14). This production can be attributed to the slow degradation of accumulated organic matter and microbial decay products. The flow rate declined gradually, confirming that the reactor was no longer being actively fed and that the remaining substrate was limited.

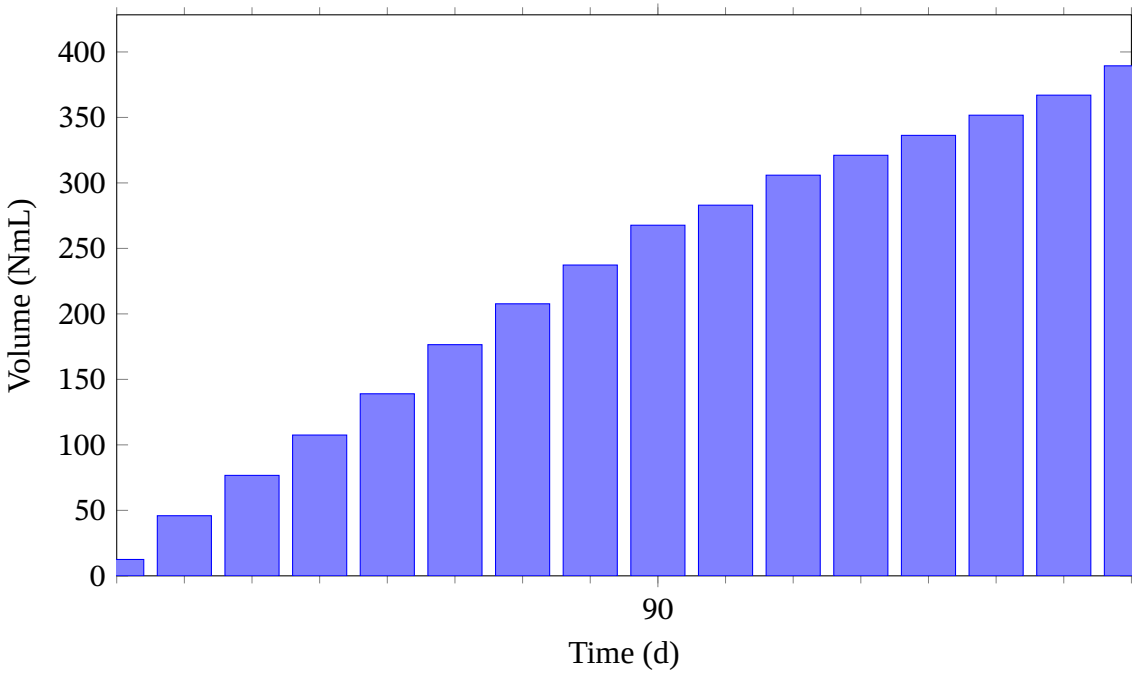


Figure 5.13: Cumulative volume of CH_4 NmL (Post-feeding period)

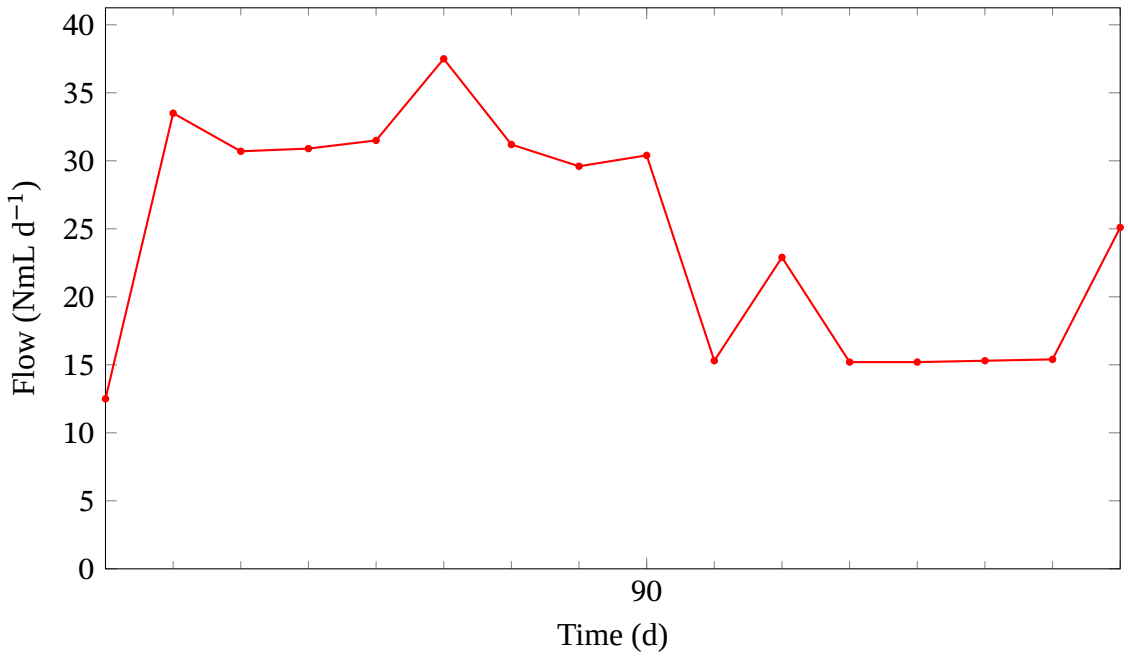


Figure 5.14: Volume flow of CH_4 NmL d^{-1} (Post-feeding period)

5.2.6 Alkalinity and Volatile Fatty Acids

Total Alk (TA) and VFA concentrations were monitored throughout the experiment to assess process stability. The results are shown in Fig. 5.15. Initially, both TA and VFA were low, reflecting the early adaptation phase before buffer addition. After the manual introduction of $KHCO_3$, TA increased sharply to values above 7,000 mg $CaCO_3$ L^{-1} and remained relatively high until close to the end. This high TA was maintained by the addition of bicarbonate to the feed, which helped to counteract the acidity produced during the degradation of the substrate.

Volatile fatty acid concentrations also increased after the start of feeding, reaching values between 3,000 mg CH_3COOH L^{-1} to 5,000 mg CH_3COOH L^{-1}. Such high VFA levels are often associated with process imbalance, as they can inhibit methanogenesis. However, the high TA provided a strong buffering capacity, and the pH remained above 6.5 for most of the experiment. With co-digestion, the pH fell progressively, and with C-GL based feed, the pH increased progressively. The VFA/TA is a more reliable indicator of instability than pH (Fig. 5.16). In this experiment, the ratio fluctuated between 0.3 to 1.2. While using the co-digestion feed, a ratio around 0.8 to 1.0 had the highest production.

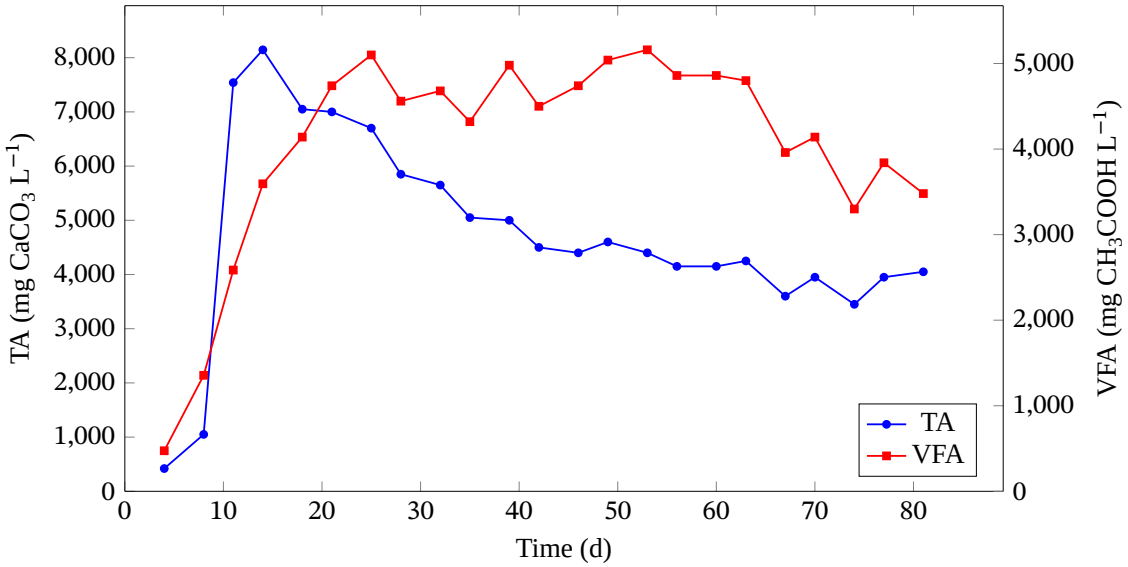


Figure 5.15: Alkalinity & volatile fatty acid comparison graph

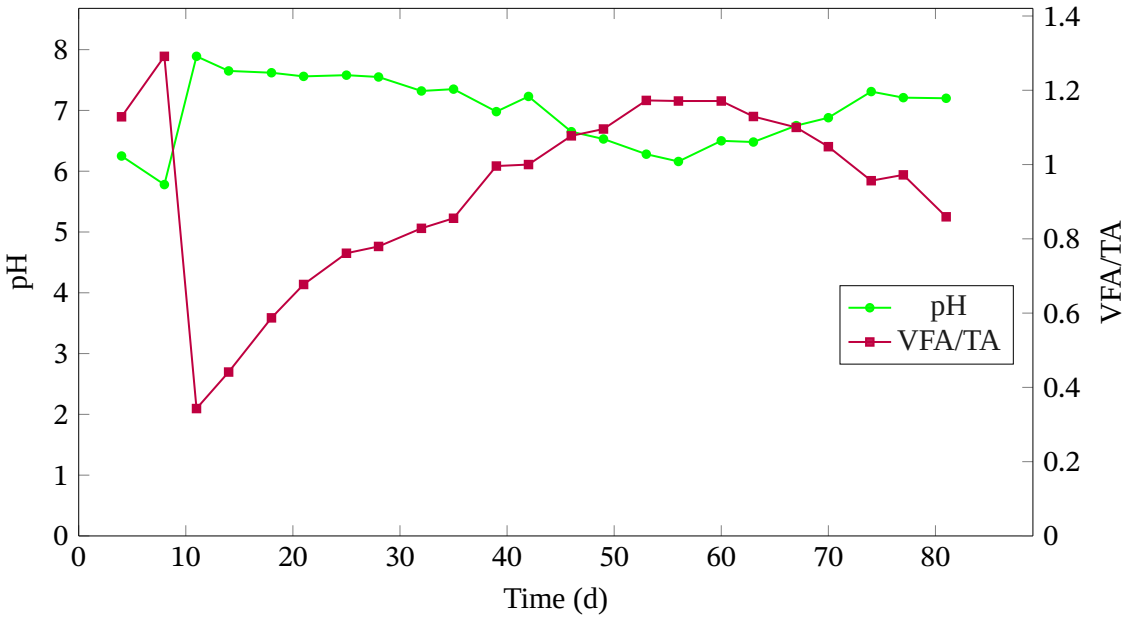


Figure 5.16: pH & VFA/TA comparison graph

Chapter 6

Discussion

The CH_4 yield data obtained throughout the experimental periods are summarised in Table 6.1, which reports the cumulative CH_4 production, specific yield, and volumetric productivity for each of the four operational phases. The cumulative specific CH_4 yield was calculated using Eq. (4.11), whereas the daily specific CH_4 production rate was obtained using Eq. (4.12). The volumetric productivity was derived from Eq. (4.14), and the phase-averaged values were computed using Eq. (4.18) to (4.20). The overall yield for the entire experiment is given by Eq. (4.21). These parameters are shown in Figs. 6.1 to 6.3, which show the evolution of CH_4 yield and OLR across the four phases.

Table 6.1: Summary of the performance per period of the experiment

Phase	Days	OLR range ($g\ VS\ L^{-1}\ d^{-1}$)	VS (g)	CH_4 (NmL)	Yield ($NmL\ g^{-1}\ VS_{fed}$)	Rate ($L\ V_r^{-1}\ d^{-1}$)
Adaptation	0 to 24	2.560e−2 to 1.500	1.663e1	7.589e2	4.564e1	3.040e−2
Co-digestion	25 to 52	1.400 to 3.160	6.342e1	1.250e3	1.971e1	4.460e−2
C-GL only	53 to 81	4.200e−1 to 1.370	2.349e1	8.779e2	3.737e1	3.030e−2
Post-feeding	82 to 97	0.000 to 1.370	9.590	3.894e2	4.060e1	2.450e−2

The overall cumulative CH_4 production was 3,276 NmL CH_4, corresponding to an overall yield of approximately 29 NmL CH_4 g^{-1} VS_{fed} (calculated using Eq. (4.21)). This value is considerably lower than those reported in the literature for similar substrates. Gebreegziabher *et al.* (2025) reviewed the AD of biodiesel byproducts and reported CH_4 yield enhancements ranging from 14 % to 226 % when C-GL is co-digested with various substrates. Tomczak *et al.* (2025) analysed data from multiple studies and concluded that

CH_4 content in biogas exceeds 60 % when the C-GL fraction in the feedstock is kept below 8 % (v/v). In the present study, the C-GL fraction varied across phases, but the overall CH_4 yield remained below the values typically reported for C-GL co-digestion. This discrepancy may be attributed to the single-stage configuration, specific characteristics of the FW and GL used, and process instability observed at the end of the co-digestion phase and the start of the GL-only phase.

During the adaptation phase (days 0 to 24), the specific CH_4 yield reached 45.64 NmL CH_4 g^{-1} VS_{fed}, the highest among all phases. However, this value must be interpreted with caution, as the cumulative VS fed during this period was only 16.628 g, and CH_4 production was still in the early stages of development. The volumetric productivity was 30.4 mL CH_4 V_r^{-1} d^{-1}. The daily yield profile in Fig. 6.2 shows a gradual increase in CH_4 production during this phase, with a notable spike around day 8, corresponding to the first addition of C-GL to the feed. This observation is consistent with Gebreegziabher *et al.* (2025), who noted that C-GL is readily biodegradable and can stimulate methanogenic activity when introduced to an acclimated microbial community.

In the co-digestion phase (days 25 to 52), the OLR was increased from 1.4 g VS L^{-1} d^{-1} to 3.16 g VS L^{-1} d^{-1}, and the cumulative CH_4 production reached 1,249.9 NmL. The specific yield decreased to 19.71 NmL CH_4 g^{-1} VS_{fed}, while the volumetric productivity increased to 0.044,6 L V_r^{-1} d^{-1}, the highest of all phases. This inverse relationship between specific yield and volumetric productivity reflects the higher organic loading rate applied, which promoted greater CH_4 production per reactor volume but reduced the efficiency of substrate conversion to CH_4. The daily yield profile in Fig. 6.2 exhibits fluctuations during this phase, which is typical of readily degradable substrates and has been reported by Tomczak *et al.* (2025) and Srimachai *et al.* (2025). Srimachai *et al.* (2025) investigated the co-digestion of canned seafood wastewater with GL waste and reported CH_4 yields of 345 mL CH_4 g^{-1} COD at a mixing ratio of 96.4 % (v/v). Although the substrate characteristics differ, the general trend of enhanced CH_4 production during co-digestion is consistent with the present findings.

The C-GL based phase (days 53 to 81) exhibited a specific yield of 37.37 NmL CH_4 g^{-1} VS_{fed} and a volumetric productivity of 0.030,3 L CH_4 V_r^{-1} d^{-1}, with a cumulative CH_4 production of 877.9 NmL CH_4. The OLR ranged from 0.42 g VS L^{-1} d^{-1} to 1.37 g VS L^{-1} d^{-1}. Despite the absence of FW in the feed, CH_4 production continued, indicating that the microbial community had adapted to GL as a sole carbon source. However, the specific yield was lower than that during the adaptation phase, suggesting that GL alone may not provide an optimal nutrient balance for methanogenesis. Tomczak *et al.* (2025) emphasised

that C-GL is an ideal co-substrate but can cause inhibition when used alone or at high concentrations due to the accumulation of VFAs and the presence of impurities such as methanol, soaps, and heavy metals. The daily yield profile in Fig. 6.2 shows a declining trend during this phase, with a notable outlier peak at day 53, corresponding to the transition from co-digestion to GL-only feeding. This anomaly, also visible in Fig. 6.3, was caused by problems during the change.

The post-feeding phase (days 82 to 97) produced 389.4 NmL of CH_4, with a specific yield of 40.60 NmL CH_4 g^{-1} VS_{fed} and a volumetric productivity of 0.024,5 L CH_4 V_r^{-1} d^{-1}. The OLR reached zero during this period, and the CH_4 produced can be attributed to the slow degradation of residual organic matter and microbial decay products. The daily yield profile in Fig. 6.2 shows a gradual decline, confirming that the reactor was no longer being actively fed and that the remaining substrate was limited.

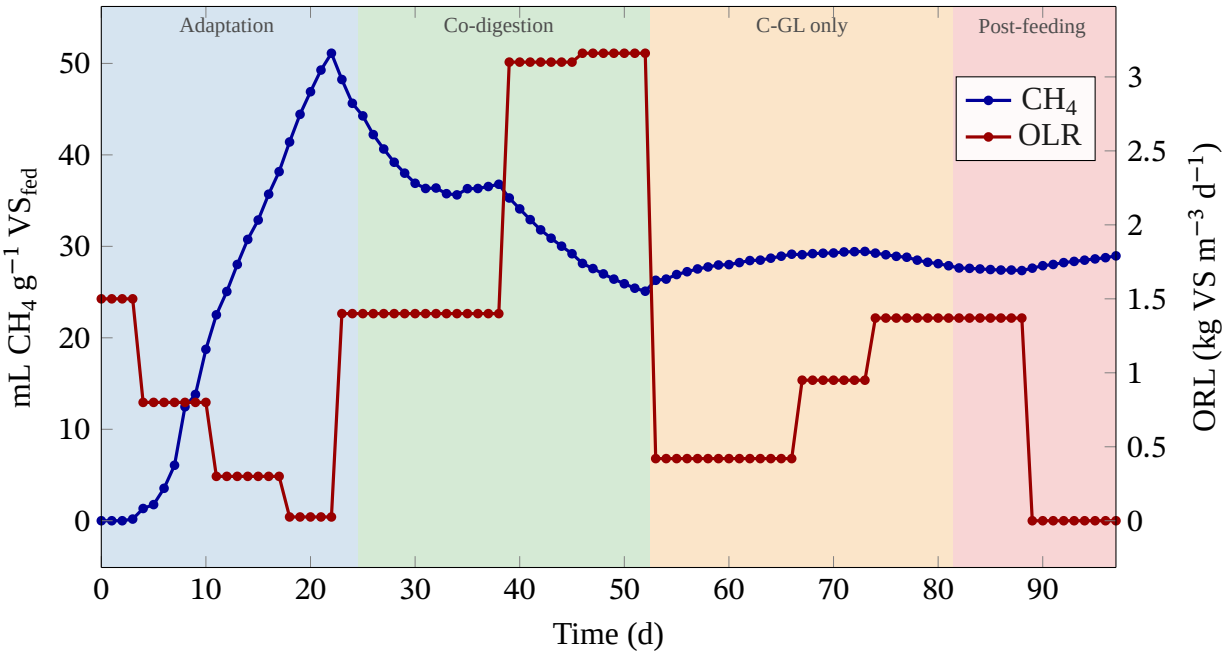


Figure 6.1: Cumulative specific CH_4 yield

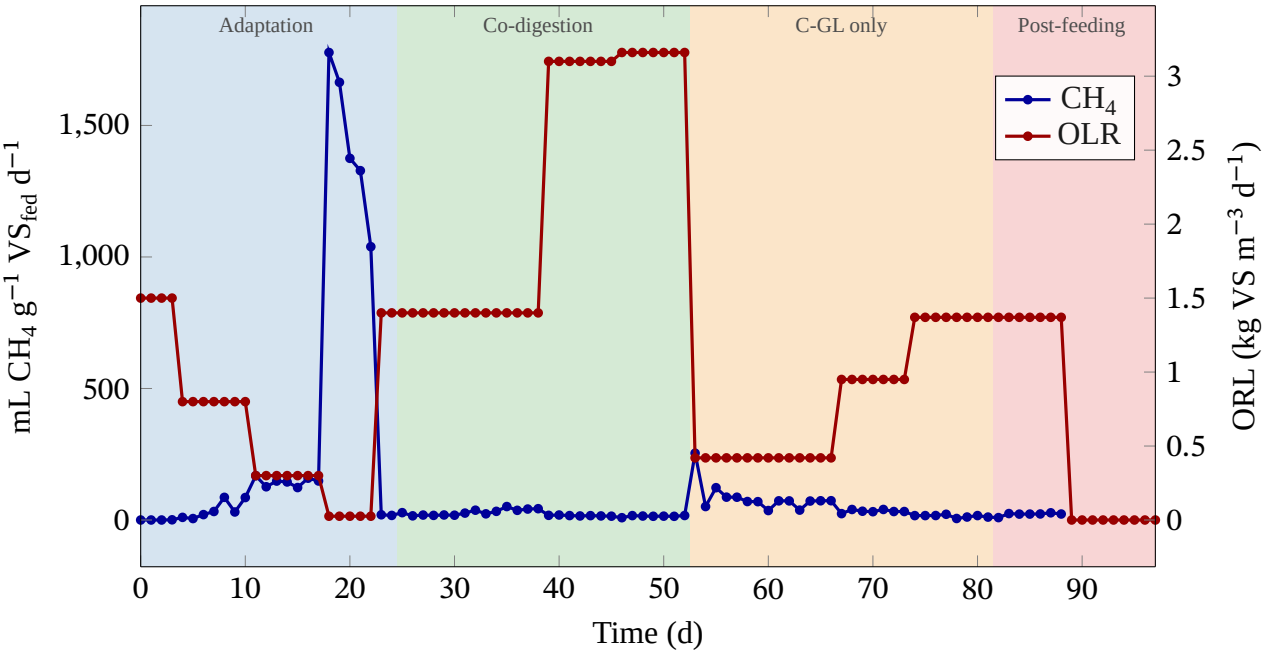


Figure 6.2: Daily yield of CH_4

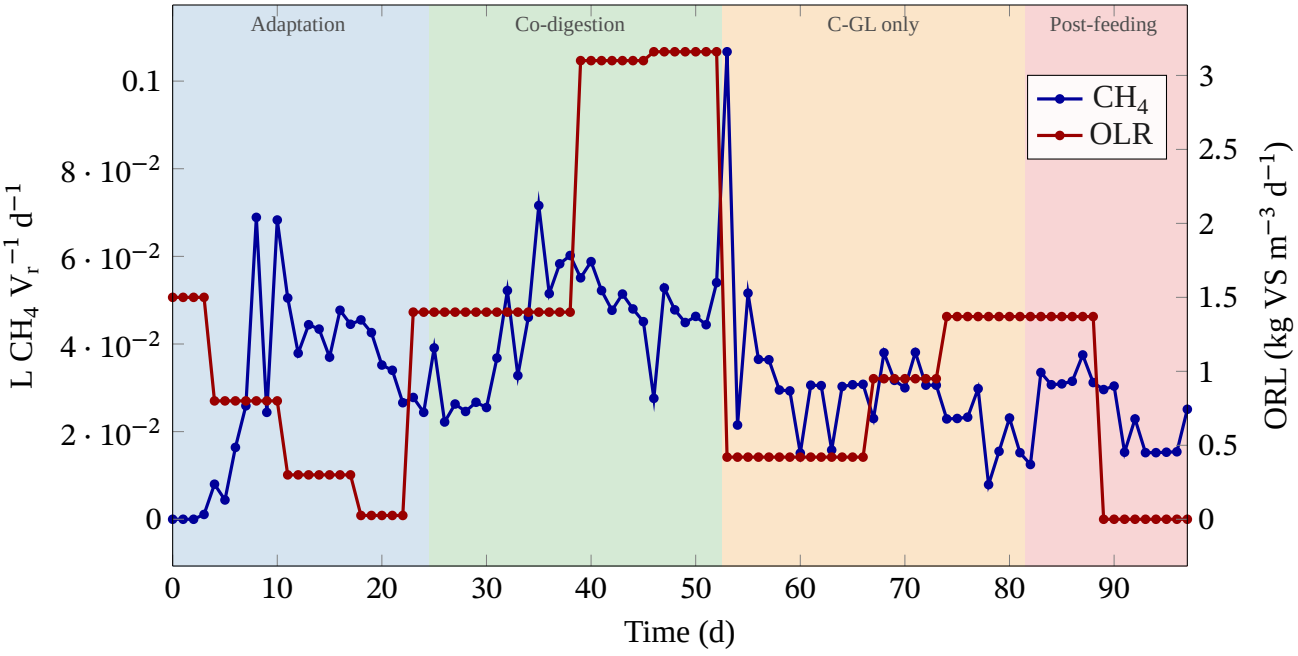


Figure 6.3: Volumetric productivity of CH_4

Comparisons with the literature revealed that the CH_4 yields obtained in this study were lower than those reported for similar co-digestion systems. Aguiar *et al.* (2026)

investigated the co-digestion of sugarcane vinasse and cattle manure in a semi-continuous stirred-tank reactor and reported volumetric CH_4 productivities of 3,981 NmL CH_4 L^{-1} d^{-1} to 5,269 NmL CH_4 L^{-1} d^{-1} for mixing ratios of 5:1 and 1:1 (v/m), respectively, at OLRs of 1.7 kg VS m^{-3} d^{-1} to 4.1 kg VS m^{-3} d^{-1}. These values are two orders of magnitude higher than the maximum volumetric productivity of 44.6 NmL CH_4 L^{-1} d^{-1} obtained in the present study. The large difference is primarily due to the much higher OLR applied by Aguiar *et al.* (2026) and the different substrate characteristics of the two studies. Aguiar *et al.* (2026) also reported that the addition of GL as a supplementary organic substrate compromised system stability, which is consistent with the VFA accumulation and pH fluctuations observed in the present study during the co-digestion phase.

Longanesi (2016) reported CH_4 yields from grape pomace digestion ranging from 20.4 NL CH_4 kg^{-1} VS in batch mode to 160 NL CH_4 kg^{-1} VS in continuous mode at an OLR of 0.4 g VS L^{-1} d^{-1}. The continuous-mode yield of 160 NL CH_4 kg^{-1} VS was equivalent to 160 NmL CH_4 g^{-1} VS, which was higher than the overall yield of 28.96 NmL CH_4 g^{-1} VS_{fed} obtained in this experiment. However, it should be noted that Longanesi (2016) used grape pomace as the sole substrate, whereas the present study used a mixture of FW and C-GL. Longanesi (2016) also reported that the co-digestion of grape pomace with maize silage improved the CH_4 yield by 113.5 NL CH_4 kg^{-1} VS, highlighting the benefits of co-digestion for balancing the C/N ratio and diluting inhibitory compounds.

The lower CH_4 yields obtained in this study can be attributed to several factors. First, the single-stage configuration may have limited the hydrolysis and acidogenesis of the FW, as the optimal conditions for these processes differ from those for methanogenesis. Srimachai *et al.* (2025) demonstrated that a two-stage system (STR - tubular reactor) can enhance CH_4 production by separating the hydrogen-producing and CH_4-producing phases. Second, the C-GL impurities may have inhibited methanogenic activity. Anand and Saxena (2012) reported that C-GL from biodiesel production contains impurities, such as methanol, soaps, and free fatty acids, that can inhibit microbial growth and fermentation. Third, the VFA/TA fluctuated from 0.3 to 1.2, indicating potential process instability. Ribas, Moraes, and Foresti (2007) emphasised the importance of accurate VFA and Alk measurements for timely detection of process imbalance. Fourth, fungal contamination observed during reactor cleaning may have competed with methanogens for substrate, potentially reducing CH_4 production. Degueurce *et al.* (2020) reported that storage of FW can lead to the growth of fungi and other microorganisms that consume organic matter and produce inhibitory compounds.

Despite the lower yields, the results demonstrate that CH_4 production was maintained

throughout the experimental period, confirming the technical feasibility of co-digesting FW and C-GL in a single-stage system. The specific CH_4 yield during the co-digestion phase (19.71 NmL CH_4 g^{-1} VS_{fed}) was lower than that during the adaptation phase, but the volumetric productivity was higher, indicating that the reactor was able to handle increased OLRs. These findings are consistent with those of Gebreegziabher *et al.* (2025), who reported that co-digestion of C-GL with nitrogen-rich substrates, such as FW, can enhance CH_4 production by balancing the C/N ratio and providing essential nutrients. However, the accumulation of VFAs and decrease in pH observed during the co-digestion phase highlight the need for careful monitoring and control of the OLR. Tomczak *et al.* (2025) recommended that the C-GL content in the feedstock should not exceed 8 % (v/v) to avoid inhibition, and that the OLR should be adjusted based on the buffering capacity of the system.

Chapter 7

Conclusion

A direct comparison of the co-digestion and C-GL based phases revealed that the co-digestion of FW and C-GL resulted in higher CH_4 yields and more stable operation. The pH decreased faster with the co-digestion, while with C-GL based phase it did not decrease that fast, and even grew with the $KHCO_3$ that was added to maintain the pH. These results support the hypothesis that C-GL, when co-digested with FW, can enhance biogas production by balancing the C/N and providing readily available carbon, whereas C-GL alone may work, but would need a different setup and ajustments.

The C-GL based phase, should not be considered strictly worse than the co-digestion of FW and C-GL, since both phases were performed in sequence, in the same reactor, and this could have given the C-GL phase disadvantages in a contaminated or exhausted environment.

Using C-GL, with some P-GL to make the necessary adjustment to balance the C/N, could be a viable solution for the C-GL problem. Its only major problem is that C-GL contains a lot of impurities that can affect anaerobic digestion, and for a system that can reliably consume it, it will need better control of its substrate.

During the entire process, the highest production of CH_4 was observed when VFA/TA was ≈1 at a pH of ≈6.5, during the C-GL based phase, even with not optimal values of pH and VFA/TA, it was able to maintain a reasonable amount of production of CH_4, showing that it can be a suitable substrate for AD.

The CH_4 yield obtained (28.96 NmL CH_4 g^{-1} VS_{fed}) was lower than the values reported in the literature for similar co-digestion systems. The highest volumetric productivity (44.6 mL $V_r{}^{-1}$ d^{-1}) was observed during the co-digestion phase, while the highest specific yield (45.64 NmL CH_4 g^{-1} VS_{fed}) was recorded during the adaptation phase. This shows the importance of controlling the OLR, VFA/TA and amount of C-GL used to enhance

CH_4 production.

Future studies could investigate the use of C-GL as a feed to mitigate its disadvantages, such as impurities and the need for an extra carbon-rich substrate to balance the feed.

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